

Hammad Khan

Blockchain Developer | Backend Developer

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 Github  Portfolio

PROFESSIONAL SUMMARY

Blockchain and Backend Developer with a BS in Software Engineering from Islamia College University, Peshawar (2024). Experienced in Solidity, Node.js, Express.js, MongoDB, PostgreSQL, Redis, REST APIs, and Web3 tools including Hardhat, ethers.js, Web3.js, and OpenZeppelin. Built smart contracts, backend APIs, authentication systems, inventory and attendance solutions, and payment integrations. Interested in Blockchain and Backend Developer roles where I can contribute to real-world software products.

WORK EXPERIENCE

- Blockchain Developer Intern, Octalooop Technologies**  01/2025 – 03/2025
Lahore, Pakistan
- Developed and tested Solidity smart contracts for token, escrow, and blockchain-based applications.
 - Integrated smart contracts with Node.js applications using ethers.js and Web3.js.
 - Used Hardhat, Mocha, and Chai for smart contract deployment, testing, and debugging.
- Backend Developer, SAT Software House**  08/2025 – 08/2026
Swabi, Pakistan
- Developed backend applications and REST APIs using Node.js, Express.js, and NestJS, handling authentication, business logic, and database operations.
 - Implemented JWT authentication, role-based access control, input validation, middleware, and data management using MongoDB, PostgreSQL, Prisma, Sequelize, and Redis.
 - Integrated third-party services and backend features such as background jobs, payment APIs, reporting, and device integrations, while coordinating API integration with the frontend team.

EDUCATION

- Bachelor of Science in Software Engineering,** 09/2020 – 08/2024
Islamia College University, Peshawar, Pakistan  Peshawar, Pakistan
- Relevant Coursework: Data Structures and Algorithms, Cryptography, Distributed Systems, Database Systems, Web Engineering, Object-Oriented Programming, Software Project Management

TECHNICAL SKILLS

- Blockchain Development** — Proficient
Solidity, Ethereum, BSC, Smart Contracts, ERC-20, ERC-721, ERC-1155, OpenZeppelin, ethers.js, Web3.js, Hardhat, Truffle, Ganache
- Databases & Tools**
MongoDB, Mongoose, PostgreSQL, Sequelize, Prisma, Redis, Git, GitHub, Postman, Swagger
- Backend Development** — Proficient
Node.js, Express.js, TypeScript, REST APIs, JWT, RBAC, Middleware, API Integration
- Testing** — Proficient
 - Mocha, Chai, Hardhat Testing

LANGUAGES

English — Proficient

Urdu — Fluent

Pashto — Native/Bilingual

PROJECTS

Attendance Management System

Technologies: NestJS, TypeScript, Prisma, Redis

- Built a backend system for managing attendance, biometric device data, background jobs, and report generation.

Retail ERP & Inventory Management System

Technologies: Node.js, SQLite, Electron.JS, Chart.js, PDFKit (invoices), SheetJS (Excel export)

- Developed backend logic for product, inventory, POS, order, and customer management, with invoice generation, Excel export, and daily backups.

Get Hire Expert – Multi-Role Backend & Payment Integration

Technologies: Node.js, Express.js, MongoDB, JWT, Axios, HMAC-SHA256

- Built backend services for Customer, Seller, and Admin modules, along with secure payment APIs, callbacks, and transaction verification.

Restaurant Management System

Technologies: Node.js, Express.js, MongoDB, Mongoose, JWT

- Developed REST APIs for user authentication, role-based access, menu management, and restaurant data handling.

Decentralised Voting System – Final Year Project,

Technologies: Solidity, Web3.js, React.js, Truffle, Ganache, Node.js, Express.js

- Built a blockchain-based voting system for voter registration, vote casting, candidate management, and on-chain results.

Qryptum Blockchain – Proof of Data (PoD) Consensus,

Technologies: Solidity, AI Validation, DRC-20 Smart Contracts, Node.js

- Developed an experimental blockchain prototype using a custom Proof of Data approach for storing, validating, and verifying data on the network.

NFT Marketplace – Auction and Direct Sales Platform,

Technologies: Solidity, ERC-721, ethers.js, Node.js, Express.js, MetaMask

- Developed smart contracts for NFT minting, listings, direct sales, auctions, and secure ownership transfers.

Escrow Marketplace – Decentralised Escrow Smart Contracts,

Technologies: Solidity, Hardhat, JavaScript, Node.js, Mocha, Chai, Git

- Built an escrow smart contract that securely manages buyer-seller transactions and fund release without a third-party intermediary.